DATA ANALYST TEST - EXECUTIVE SUMMARY

Candidate: Mahmoud Abdallah
Date: August 1, 2026
Test: Consumer Banking Product Research
Institution: WalletHub HR Recruitment

PROJECT OVERVIEW

Completed comprehensive consumer banking product research across six financial institutions as part of the Data Analyst test for WalletHub. The project involved collecting, validating, and analyzing financial product data from multiple sources while documenting methodologies and limitations.

SCOPE OF WORK

Institutions Researched:

1. OneAZ Credit Union (oneazcu.com)
2. Wikipedia (compound interest formula verification)
3. Capital One (capitalone.com)
4. Wells Fargo (wellsfargo.com)
5. First American Bank (firstambank.com)
6. Union Savings Bank (unionsavings.com)
7. BMO (bmo.com)

Questions Addressed:

- Consumer product offerings and categorization
- Interest rate structures and APY calculations
- Fee structures and account requirements
- Government-insured deposit products

- Fixed vs. variable rate product analysis
- Overdraft fee policies

METHODOLOGY

Data Collection:

- Direct verification from official institutional websites
- Cross-referencing with rate disclosure pages and fee schedules
- Utilization of third-party financial rate trackers for validation
- Systematic documentation of all source URLs

Data Validation:

- Mathematical validation of APY calculations using standard compound interest formula
- Cross-verification of rate information across multiple pages
- Confirmation of fee structures and account requirements
- Validation of insurance coverage details

Analysis Approach:

- Standardized metrics across all six institutions
- Categorization of products by type and rate structure
- Distinction between dynamic variable-rate and fixed-rate products
- Documentation of all discrepancies and limitations

KEY ACHIEVEMENTS

1. Comprehensive Product Research

- Successfully identified and categorized all consumer products for OneAZ Credit Union
- Documented complete product suites including banking, lending, and investment services
- Verified auto loan availability and terms for new and used vehicles

2. Accurate Rate Analysis

- Retrieved and documented Premium Checking Account tiered rates (0.10%-0.20% APY)
- Validated Capital One Money Market flat rate structure (1.00% APY)
- Confirmed Union Savings Bank fixed-rate products (CDs, IRA CDs)

3. Mathematical Precision

- Correctly calculated APY using compound interest formula: **APY = (1 + i_nom/N)^N - 1**
- Applied formula with inputs: i_nom = 0.0134, N = 365
- Derived accurate result: **1.3490% APY** (~1.35%)

4. Fee Structure Documentation

- Detailed Wells Fargo Way2Save account requirements and restrictions
- Documented BMO overdraft fee limits (maximum 3 per business day)
- Identified fee waiver conditions across multiple products

5. Government Insurance Verification

- Confirmed FDIC insurance coverage for First American Bank deposit products
- Listed all insured account types: checking, savings, money market, CDs
- Verified NCUA insurance for OneAZ Credit Union

RESPONSE TO FEEDBACK

Original Feedback from WalletHub HR (Lotika Lawrence):

> *"You did well but there is room for improvement so we will give you the chance to review and improve some of your answers. Please take another look at your solutions for the DA test (for questions 1a, 1c, 3a, 4a and 4b in particular)."*

Improvements Made:

Question 1a (OneAZ Products):

- Expanded from brief list to comprehensive categorization
- Added specific product names and features

- Included rate ranges and special programs
- Organized by product type for clarity

Question 1c (Premium Checking Rates):

- Provided exact tiered rate structure with balance ranges
- Included dividend rate, APY, and interest rate for each tier
- Added minimum balance requirements and fee information
- Specified source URL for verification

Question 3a (Capital One Money Market):

- Retrieved official rate disclosure document
- Documented exact interest rate (0.9954%) and APY (1.00%)
- Clarified flat rate structure regardless of balance
- Added compounding and crediting frequency

Question 4a (First American FDIC):

- Located official FDIC insurance policy page
- Listed all four insured deposit account types
- Added FDIC backing details and coverage limits
- Included historical context for credibility

Question 4b (Union Savings Fixed Rates):

- Identified all fixed-rate products (Flex CD, Norwalk CD Special, IRA CDs, Standard CDs)
- Explained rate stability characteristics
- Added FDIC insurance confirmation
- Specified term lengths and features

DELIVERABLES

Documents Submitted:

1. **Original Word Document:** Initial test responses
2. **Summary PDF:** Executive summary of findings
3. **Complete Answer Document:** Detailed responses to all questions
4. **This Summary:** Professional overview of methodology and achievements

Data Quality:

- All information verified from official sources
- Rates current as of July-August 2026
- Source URLs provided for each data point
- Mathematical calculations validated

SKILLS DEMONSTRATED

Research Skills:

Web research and data collection
Source verification and cross-referencing
Financial product knowledge
Institutional website navigation

Analytical Skills:

Financial rate analysis
APY calculation and validation
Fee structure interpretation
Product categorization

Technical Skills:

Mathematical formula application
Data standardization
Documentation and reporting
Attention to detail

Professional Skills:

Clear written communication (B2+ English)
Professional document formatting
Response to feedback
Timely delivery

CHALLENGES OVERCOME

1. **Missing Public Data:** Some rate information required accessing disclosure documents rather than public pages
2. **Variable Rates:** Distinguishing between fixed and variable rate products across institutions
3. **Institutional Variations:** Standardizing metrics across different types of financial institutions (banks vs. credit unions)
4. **Mathematical Precision:** Ensuring accurate APY calculations with proper formula application

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LESSONS LEARNED

1. Always check official disclosure documents for the most accurate rate information
2. Distinguish between nominal rates, dividend rates, and APY for different institution types
3. Verify fee structures and waiver conditions directly from source
4. Document all source URLs for transparency and verification
5. Cross-reference information across multiple pages for accuracy

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CONCLUSION

This project demonstrates the ability to:

- Collect and validate financial data from multiple sources
- Apply mathematical formulas accurately
- Organize and present information professionally
- Respond to feedback and improve deliverables
- Maintain attention to detail and data accuracy

All test questions have been addressed with comprehensive, verified answers suitable for professional submission.

Prepared by: Mahmoud Abdallah
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Status: Complete and Ready for Submission